
OFFICE	Weinstein Computer Science 116 Florida Southern College Lakeland, FL 33801 USA	E-mail: hngo@flsouthern.edu Total number of citations: > 1,400 Citation h-index: 19
RESEARCH INTERESTS	Artificial Intelligence for Health, Cancer and Neurodegenerative Diseases Diagnostics, Digital Health, Computational Bioengineering, Diagnostics in Low-Resource Settings	
TRAINING	Johns Hopkins University , Baltimore, Maryland USA May, 2021 - July, 2023 Postdoctoral Scholar Advisor: Jeff "Tza-Huei" Wang, Ph.D. Duke Eye Center , Durham, North Carolina USA April, 2017 - December, 2017 Postdoctoral Scholar Advisor: Joseph Izatt, Ph.D. and Cynthia Toth, MD	
EDUCATION	Duke University , Durham, North Carolina USA August, 2011 - March, 2017 Ph.D. in Biomedical Engineering, May, 2017 Advisor: Tuan Vo-Dinh, Ph.D. Korea Maritime University , Busan, South Korea August, 2006 - August, 2008 M.E, Control and Instrument Engineering, August, 2008 University of Technology , Ho Chi Minh City, Vietnam September, 2001 – April, 2006 B.E, Mechatronics, April, 2006	
RESEARCH GRANTS	• August, 2024 - July, 2027: Adapting a machine learning algorithm to predict thyroid cytopathology in LMIC. Sponsor: National Institutes of Health (National Cancer Institute). Award number: # 4R33CA268428-03 (3 years, \$287,551/year). Co-PI. • January, 2025 - January, 2028: The application of Artificial Intelligence (AI) in the diagnosis and prognosis of Alzheimer's disease. Sponsor: Vietnamese Ministry of Science and Technology. Program KC.10/21-30 (approx. \$284,437). Co-I.	
SELECTED PUBLICATIONS	Below are 5 selected publications. For full list, refer to my Google Scholar 1. Aiden Coffey, Vladimir Cuc, Hoan Thanh Ngo , Walter Lee. <i>Thyroid Cancer Detection using Smartphone-captured Cytopathology Images and Foundation</i>	

Model (Accepted for Medical Imaging with Deep Learning (MIDL), Salt Lake City, Utah, United States, Jul 09 2025).

2. Quan Thanh Huynh, Hieu Xuan Le, Lien Tran, Trung Nguyen, Luan Nguyen, Alex Mremi, Daniel Mbwapbo, Walter Lee, **Hoan Thanh Ngo**. *Explainable AI for Thyroid Cancer Detection using Smartphone-captured Cytopathology Images and Multi-Instance Learning* (Accepted for and presented at BMES 2024 Annual Meeting Oct 23-26, Baltimore, MD USA).
3. **Hoan T Ngo**, Patarajarin Akarapipad, Pei-Wei Lee, Joon Soo Park, Fan-En Chen, Alexander Y Trick, Tza-Huei Wang, Kuangwen Hsieh. *Rapid and portable quantification of HIV RNA via a smartphone-enabled digital CRISPR device and deep learning*. Sensors and Actuators Reports, 8, 100212, 2024.
4. Dong Nguyen, Hoang Nguyen, Hong Ong, Hoang Le, Huong Ha, Nguyen Thanh Duc, **Hoan T Ngo**. *Ensemble learning using traditional machine learning and deep neural network for diagnosis of Alzheimer's Disease*. IBRO Neuroscience Reports, 13, 255-263, 2022.
5. Quan Thanh Huynh, Phuc Hoang Nguyen, Hieu Xuan Le, Lua Thi Ngo, Nhu-Thuy Trinh, Mai Thi-Thanh Tran, Hoan Tam Nguyen, Nga Thi Vu, Anh Tam Nguyen, Kazuma Suda, Kazuhiro Tsuji, Tsuyoshi Ishii, Trung Xuan Ngo, and **Hoan T Ngo**. *Automatic acne object detection and acne severity grading using smartphone images and artificial intelligence*. Diagnostics, 12(8), 1879, 2022.

BOOK/BOOK CHAPTER PUBLICATIONS

1. **Hoan T Ngo**, Tuan Vo-Dinh, "SERS biosensors for Infectious Disease Diagnosis". Chapter 3 in the Book SERS for Point-of-care and Clinical Applications. Elsevier Science, 2022.
2. Van Toi, V., Le, T.Q., **Ngo, H.T.**, Nguyen, T.-H. (Eds.), Book "7th International Conference on the Development of Biomedical Engineering in Vietnam (BME7)", IFMBE Proceedings, Springer Singapore publisher, 2019.
3. **HT Ngo**, N Gandra, AM Fales, SM Taylor, T Vo-Dinh, "Sensitive DNA Detection and SNP Identification Using Ultrabright SERS Nanorattles and Magnetic Beads for In Vitro Diagnostics", Chapter 27 in the Book Nanotechnology in Biology and Medicine: Methods, Devices, and Applications, Second Edition, Taylor and Francis publisher, 2017.
4. T. Vo-Dinh, HN Wang, **HT Ngo**, "SERS Molecular Sentinel Nanoprobes to Detect Biomarkers for Medical Diagnostics", Chapter 24 in the Book Biomedical Photonics Handbook, Second Edition: Therapeutics and Advanced Biophotonics, Taylor and Francis publisher, 2015.

PATENTS

1. T Vo-Dinh, N Gandra and **HT Ngo**, "Nanoprobe compositions and methods of use thereof", US Patent 10,876,150, 2020.

EMPLOYMENT

Assistant Professor

August, 2023 - present

Department of Computer Science & Engineering

Florida Southern College, Lakeland, Florida USA

- Developing an AI-powered system for thyroid cancer detection in low-resource settings (in collaboration with Walter Lee, MD at Duke University School of Medicine and medical doctors from Tanzania and Vietnam).
- Developing an AI-powered system for early Alzheimer's disease detection using retinal images and brain MRI images (in collaboration with medical doctors in Vietnam).
- Developing an AI-powered system for psoriasis severity indexing using smartphone images (in collaboration with Scott Elman, MD at University of Miami Miller School of Medicine).

Postdoctoral Scholar

May, 2021 - July, 2023

Department of Biomedical Engineering and Mechanical Engineering

Johns Hopkins University, Baltimore, Maryland USA

Advisor: Dr. Jeff "Tza-huei" Wang

- Developed a PCR-based method and portable device for HIV viral load quantification at the point-of-care.
- Developed a CRISPR-based method and a smartphone-based device + Deep Learning for rapid and portable HIV RNA quantification.

Faculty & Vice Chair

January, 2018 - April, 2021

Department of Biomedical Engineering

International University - Vietnam National University, HCM City, Vietnam

- Developed various low-cost medical devices for developing countries including a low-cost PCR machine, a smartphone-based fundus camera, a smart blind stick, etc.
- Developed AI algorithms for various medical image analysis.

Postdoctoral Scholar

April, 2017 - December, 2017

Duke Eye Center

Duke University, Durham, North Carolina USA

Advisor: Dr. Joseph Izatt and Cynthia Toth, MD

- Developed an optical imaging system based on Optical Coherent Tomography (OCT) for guidance of ophthalmic microsurgeries.
- Developed algorithms for analysis of OCT angiography images.

Research Assistant

August, 2011 - March, 2017

Department of Biomedical Engineering

Duke University, Durham, North Carolina USA

Advisor: Dr. Tuan Vo-Dinh

- Developed biosensors and integrated systems for point-of-care diagnostics of respiratory infection, dengue, and malaria.
- Developed biosensors and integrated systems for point-of-care diagnostics of head and neck cancer.

Founding Faculty

October, 2008 - July, 2011

Department of Biomedical Engineering & School of Electrical Engineering

International University - Vietnam National University, HCM City, Vietnam

- Developed inductive plethysmography sensor for respiratory monitoring.
- Developed an algorithm for extracting fetal ECG from maternal ECG.

TEACHING EXPERIENCE

Florida Southern College, Lakeland, Florida USA

August, 2023 - present

Department of Computer Science and Engineering

Courses taught:

- Introduction to Computer Science (Python Programming)
- Computer Organization and Architecture
- Computer Science Research I & II (topic: AI in Medicine)
- Microcontroller Programming
- Introduction to Embedded Systems

International Uni. - Vietnam National Uni., HCM City, Vietnam

January, 2018 -

April, 2021

Department of Biomedical Engineering

Courses taught:

- Medical Imaging
- Medical Instrumentation
- Medical Instrumentation Design
- Biosignal Processing
- Biomedical Image Processing
- Artificial Intelligence for Biomedical Engineers

Duke University, Durham, North Carolina USA

August, 2011 - March, 2017

Department of Biomedical Engineering

TAed:

- Biomedical Electronic Measurements I
- Advances in Photonics

International Uni. - Vietnam National Uni., HCM City, Vietnam

October, 2008 -

July, 2011

Department of Biomedical Engineering & School of Electrical Engineering

Courses taught:

- Principles of Electrical Engineering
- Control Systems

- Microcontroller Programming
- Biomedical Engineering Design

INDUSTRY EXPERIENCE	Part-time employee SonarTech Co., Ltd., Busan, South Korea August, 2006 - August, 2008 • Developed a control system for underwater robot.
FELLOWSHIPS, AWARDS, AND HONORS	• Fitzpatrick Foundation Scholars Fellowship, 2014-2016 • Best Poster Award, Duke University Fitzpatrick Institute for Photonics Symposium, 2015 • Best Paper Award from SPIE, 2014 • Vietnam Education Foundation Fellow, 2011 • Graduated Summa Cum Laude and received Silver Medal, University of Technology, Ho Chi Minh City, Vietnam, 2006 • Champion of a robot contest organized by the University of Natural Science, HCM City, 2005 • Certificate of merit from ROBOCON Vietnam 2005, the largest robotic contest in Vietnam, 2005 • Intel Vietnam scholarship, 2002
PROFESSIONAL AFFILIATIONS	• Biomedical Engineering Society (BMES) • International Society to Advance Alzheimer's Research and Treatment (ISTAART) • The International Society of Optical Engineering (SPIE)
PROFESSIONAL SERVICE	• Co-chair, Machine Learning II session, Biomedical Engineering Society (BMES) Annual Meeting, Baltimore, Maryland USA, October, 2024. • Associate Editor, Frontiers in Lab on a Chip Technologies (May, 2023 – present). • Manuscript Reviewer, Nature Communication, Sensors, etc. • Co-organizer, the 3rd (2010), 7th (2018), and 8th (2020) International Conference on the Development of Biomedical Engineering in Vietnam. • Organizer, the AI for Health workshop at International University - Vietnam National University, HCM City, Vietnam, 2019.
COMPUTING SKILLS	• Completed courses and received certificate in the Deep Learning Specialization taught by AI expert, Dr. Andrew Ng. • Completed course and received certificate in Fundamentals of Machine Learning for Healthcare (Stanford University). • TensorFlow in Practice Specialization: completed the TensorFlow in Practice Specialization taught by Laurence Moroney, an AI Lead at Google. • Python for Data Science and AI: completed IBM's Python for Data Science and AI • PyTorch, MATLAB, COMSOL Multiphysics; OpticStudio by Zemax; Mathematica for symbolic mathematical computing.

ENGINEERING
AND
LIFE-SCIENCE
SKILLS

- Device Engineering Skills: Electrical and Electronics Circuit Design; Printed Circuit Board (PCB) Design; Microcontroller Programming (Arduino, ESP32, Raspberry Pi, Raspberry Pi Pico); FPGA Programming; LabVIEW; Mechanical Design and 3D Modeling (SolidWorks, Fusion 360); Machining (CNC, Milling, Lathing); Rapid Prototyping using 3D Printing and Laser Cutting, Thermoforming; Mobile App Prototyping using MIT App Inventor 2.
- Micro/Nano Fabrication and Characterization Skills: Nanochip Fabrication; Nanoparticle Synthesis; Photolithography; Electron Beam Lithography; Plasma Asher; E-beam & Thermal Metal Evaporator; DC Sputter Coating of Metals; Sputter Coater; Reactive Ion Etcher (RIE); Plasma Enhanced Chemical Vapor Deposition System (PECVD); Scanning Electron Microscope (SEM); Transmission Electron Microscope (TEM); X-Ray Photoelectron Spectrometer (XPS); Raman and surface enhanced Raman Spectroscopy using Renishaw InVia Raman Microscope and Horiba Jobin Yvon LabRam ARAMIS; UV-Vis-NIR Spectrophotometer; Atomic Force Microscope (AFM); Near-field Scanning Optical Microscope (NSOM); Fluorescence Microscope; Olympus FV1000 Multiphoton Microscope; Optical Coherent Tomography (OCT); Clean room skills.
- Life Science Skills: Cell Culture; Cell Counting; Multi-mode Plate Reader; Nucleic Acid (DNA and RNA) Extraction; Real time PCR; Digital PCR using QuantStudio chip; Isothermal nucleic acid amplification (LAMP and RPA); CRISPR Detection; DNA Probe Design; Gel Electrophoresis; Bioconjugation.